

AI-Driven Intelligent Security Systems for Multi-Domain Threat Detection and Response

Group 5

Members:

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1. Introduction and Background

1.1 Problem Description

With the rapid growth of digital services and online platforms, organizations face increasingly complex cybersecurity threats. These threats appear in multiple domains including email communication, web applications, multimedia content, and enterprise infrastructures.

Modern cyber attacks involve various techniques such as phishing attacks, web-based attacks, identity compromise, deepfake media manipulation, and advanced persistent threats targeting enterprise systems.

Traditional security mechanisms often rely on rule-based detection systems. While these approaches are effective for known attack patterns, they struggle to detect evolving threats, behavioral anomalies, and sophisticated attack chains.

Artificial Intelligence (AI) and Machine Learning (ML) techniques have shown significant potential in improving cybersecurity defense systems. AI models can analyze large volumes of security data, detect anomalies, and identify attack patterns more effectively than traditional rule-based approaches.

However, most existing solutions focus on solving isolated security problems such as phishing detection or anomaly detection in network traffic. There is a lack of integrated intelligent security systems capable of addressing threats across multiple domains simultaneously.

Therefore, this project aims to explore AI-driven intelligent security systems that can detect and respond to cybersecurity threats across different application domains including web systems, email communication, multimedia content, and enterprise infrastructure.

1.2 Research Objectives

The primary objective of this group project is to investigate and develop AI-driven cybersecurity solutions capable of detecting threats across multiple domains.

Specific objectives include:

- Investigate existing AI techniques used in cybersecurity threat detection.
- Analyze different types of cyber threats such as phishing, web attacks, deepfake media manipulation, and infrastructure threats.
- Design intelligent security systems capable of detecting and responding to these threats.
- Evaluate the effectiveness of AI-based security solutions in different cybersecurity scenarios.

Research questions:

1. What are the current AI-based methods used for cybersecurity threat detection and what are their advantages and limitations?
2. How can AI techniques be applied to detect different categories of cyber threats including phishing, web attacks, deepfakes, and enterprise system threats?
3. How can the effectiveness of AI-based security systems be evaluated across multiple security domains?

1.3 Scope of the Project

The project focuses on investigating AI-based solutions for threat detection across multiple cybersecurity domains.

Included:

- Phishing email and fake website detection
- Behavioral authentication and web attack detection
- Multimodal deepfake detection
- Autonomous security agents for SOC monitoring

Excluded:

- Enterprise-scale deployment
- Real-time national-scale cyber defense systems
- Offensive cybersecurity testing

The project will focus on developing **prototype AI-based detection systems** in each selected domain.

2. Proposed Solution

2.1 Solution Description

The proposed project explores the development of intelligent AI-based security systems that detect and respond to cybersecurity threats across multiple domains.

Each group member focuses on a specific domain:

Nguyễn Ngọc Bách

Phishing email and fake website detection using NLP and machine learning.

Trần Văn Khôi

Behavior-based continuous authentication and web attack detection using machine learning and NLP techniques.

Phạm Quang Huy

Multimodal deepfake detection integrating biometric signals and audio-visual synchronization.

Nguyễn Đức Bình

SentinelAI – an autonomous security agent for SOC and data center threat intelligence and response.

These systems collectively demonstrate how AI technologies can enhance cybersecurity capabilities in different contexts.

2.2 Software Architecture (General)

The proposed system architecture consists of multiple AI-based detection modules operating in different domains.

Cyber Data Sources

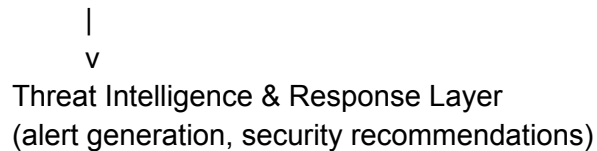
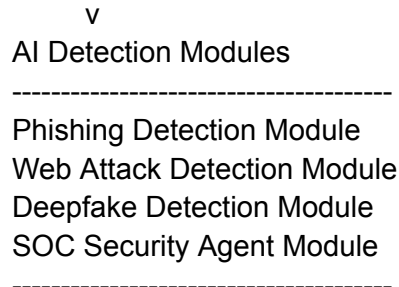
(email, web traffic, multimedia content, system logs)

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Data Processing Layer

(log collection, feature extraction)

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Each module focuses on detecting specific threats while contributing to the overall cybersecurity monitoring system.

2.3 Technology and Tools Used

The following technologies will be used:

Python – main programming language
Scikit-learn / PyTorch – machine learning frameworks
NLP libraries – text analysis for phishing detection
Computer Vision libraries – deepfake detection
FastAPI – backend services
Docker – containerized development

These technologies are widely used in AI-based cybersecurity research and support scalable development.

3. Implementation Plan

3.1 Development Stages

Phase 1 – Research and Requirement Analysis
Study cybersecurity threats and AI-based detection techniques.

Phase 2 – System Design
Design system architecture and define module responsibilities.

Phase 3 – Development

Implement AI-based detection modules for each domain.

Phase 4 – Testing and Evaluation

Test the developed systems using available datasets.

Phase 5 – Documentation

Prepare final reports and project documentation.

3.2 Expected Schedule

Month 1 – Research and analysis

Month 2 – System design

Month 3–4 – Development

Month 5 – Testing and evaluation

3.3 Feasibility Assessment

Potential challenges include limited availability of real cybersecurity datasets and difficulties in detecting sophisticated attack patterns.

These challenges can be mitigated by using publicly available datasets and simulated environments.

Required resources include development machines, open-source AI frameworks, and publicly available cybersecurity datasets.

4. Expected Results

The project is expected to deliver:

- Multiple AI-based threat detection prototypes
- Improved detection capability for phishing, web attacks, deepfakes, and infrastructure threats
- A research report describing the system design and evaluation results

5. Evaluation Plan

Evaluation metrics may include:

Detection accuracy
False positive rate
Threat detection performance
System usability

Each module will be evaluated using appropriate datasets and testing scenarios.

6. References

- [1] É. Lefèvre, "AI-Based Modeling in Software Engineering: Techniques, Applications, and Future Directions," 2022.
- [2] I. H. Sarker, "AI-Based Modeling: Techniques, Applications and Research Issues," SN Computer Science, 2022.
- [3] D. De Silva et al., "An Artificial Intelligence Life Cycle: From Conception to Production," Machine Learning with Applications, 2022.